Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/03/2023
PI Name:	Delp, Michael Ph.D.		
Project Title:	Effects of Simulated Microgravity and Partial Unloading on Organ Systems of the Body		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mdelp@fsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-214-1195
Organization Name:	Florida State University		
PI Address 1:	College of Human Sciences		
PI Address 2:	242 Sandels Building, 120 Convocation Way		
PI Web Page:			
City:	Tallahassee	**State:**	FL
Zip Code:	32306-0001	**Congressional District:**	2
Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	01/15/2020	**End Date:**	01/14/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NOTE: Per PI Report, Dr. Claire Berryman and Dr, Stephen Hennigar are no longer CoInvestigators with the project (Ed., 1/16/2023).		
COI Name (Institution):	Gordon, Bradley Ph.D. (Florida State University) Kim, Jeong-Su Ph.D. (Florida State University) Mercer, Roger Ph.D. (Florida State University) Mortreux, Marie Ph.D. (Beth Israel Deaconess Medical Center, Inc.) Singh, Prashant Ph.D. (Florida State University) Steiner, Jennifer Ph.D. (Florida State University) Allen, Matthew Ph.D. (Indiana University) Mao, Xiao Wen M.D. (Loma Linda University) Rutkove, Seward M.D. (Beth Israel Deaconess Medical Center, Inc.) Narayanan, Anand Ph.D. (Florida State University)		
Grant/Contract No.:	80NSSC19K1599		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	The human body has evolved over the millennia to a narrow range of environmental conditions on Earth. With the advent of human spaceflight, we have now begun to understand the pervasive role Earth's gravity (1-G) plays in regulating organ homeostasis. Numerous factors could initiate or contribute to tissue adaptations that occur during spaceflight, including head-ward fluid shifts, tissue unloading, physical inactivity, and environmental factors other than gravity (e.g., pCO2, pO2, temperature). Weightlessness is known to cause a redistribution of bodily fluids (blood, lymph, organ fluids) and unloading (i.e., lack of gravity pulling organs, physical movement during walking, standing, etc.). The degree of influence of systemic adaptations between bodily fluid redistribution versus unloading is unknown for most tissues. Additionally, spaceflight-induced adaptations in one organ system may have effects on the function of other organs, such as digestive dysfunction associated with systemic inflammation. Using a rat model to simulate the weightless environment, the purpose of this proposal is to 1) examine the effects of 30-d head-ward fluid shifts and unloading on the structure and function of arteries, lymphatics, digestive, bone, skeletal muscle, visual, and brain neural systems by contrasting adaptations induced by head-down unloading with horizontal unloading, and 2) to determine whether 30-d of partial horizontal unloading to simulate Moon (1/8-G) and Mars (3/8-G) gravity will attenuate or abolish adaptations induced by 30-d of full horizontal unloading. Measurements specific to the structure and function of arteries, lymphatics, digestive, bone, skeletal muscle, visual and brain neural systems will be made, as well as examining the proteome of these various organs through proteomic analysis. The results will provide important and new information regarding what role, if any, head-ward fluid shifts play in inducing tissue adaptations with simulated weightlessness, and whether these adaptations are attenuated or abolished with partial loading.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Much of what we know about the effects of spaceflight on the human body is segmented into the effects on specific organ systems, i.e., bone, muscle, or cardiovascular systems. The current projects looks simultaneously at multiple organ systems and how that interact and adapt. Such information is also important for understanding life on Earth, and how changes in one organ system of the body may affect or influence the function of other organ systems. This is one of the strengths of the present proposal.
Task Progress:	This project requires specialized cages to model Moon and Mars gravity in rats, and some specialized instrumentation. Cages and instrumentation development and fabrication have been completed. Animal experimentation is to commence soon.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Articles in Peer-reviewed Journals	Willey JS, Britten RA, Blaber E, Tahimic CGT, Chancellor J, Mortreux M, Sanford LD, Kubik AJ, Delp MD, Mao XW. "The individual and combined effects of spaceflight radiation and microgravity on biologic systems and functional outcomes." J Environ Sci Health C Toxicol Carcinog. 2021 Apr 27;39(2):129-79. https://doi.org/10.1080/26896583.2021.1885283 ; PMID: 33902391 , Apr-2021